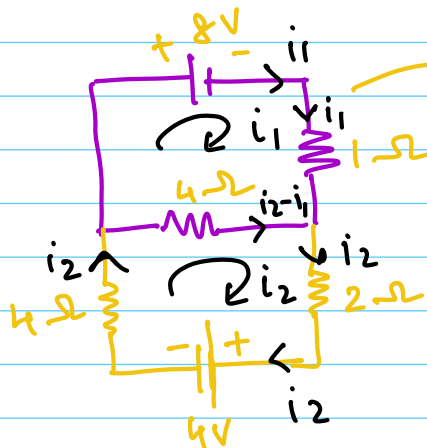


Lec-08

CIRCUITS

Linear Algebra
can be used in
analysis of
circuits.



Determine currents
in branches

i_1, i_2 MESH currents

MESH ANALYSIS.

$$\text{loop 1: } 8 + i_1 - 4(i_2 - i_1) = 0$$

$$\text{loop 2: } 4i_2 + 4(i_2 - i_1) + 2i_2 + 4i_2 = 0$$

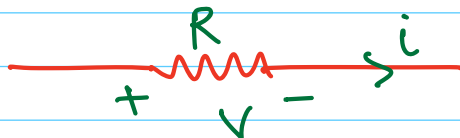
Kirchoff's voltage law

KVL :

Net voltage drop across loop is 0.

+ Ohm's law

Voltage drop across resistor



$$V = iR$$

Voltage drop in
direction of current

$$\text{Loop 1: } 8 + i_1 - 4i_2 + 4i_1 = 0$$

$$\Rightarrow 5i_1 - 4i_2 = -8$$

$$\text{Loop 2: } 4i_2 - 4i_1 + 2i_2 + 4 + 4i_2 = 0$$

$$\Rightarrow -4i_1 + 10i_2 = -4$$

$$5i_1 - 4i_2 = -8$$

$$-4i_1 + 10i_2 = -4$$

$$\Rightarrow \begin{bmatrix} 5 & -4 \\ -4 & 10 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} -8 \\ -4 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_A \quad \underbrace{\hspace{10em}}_{\bar{b}}$

$$\Rightarrow \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = A^{-1} \bar{b}$$

$$\Rightarrow \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \frac{1}{34} \begin{bmatrix} 10 & 4 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -8 \\ -4 \end{bmatrix}$$

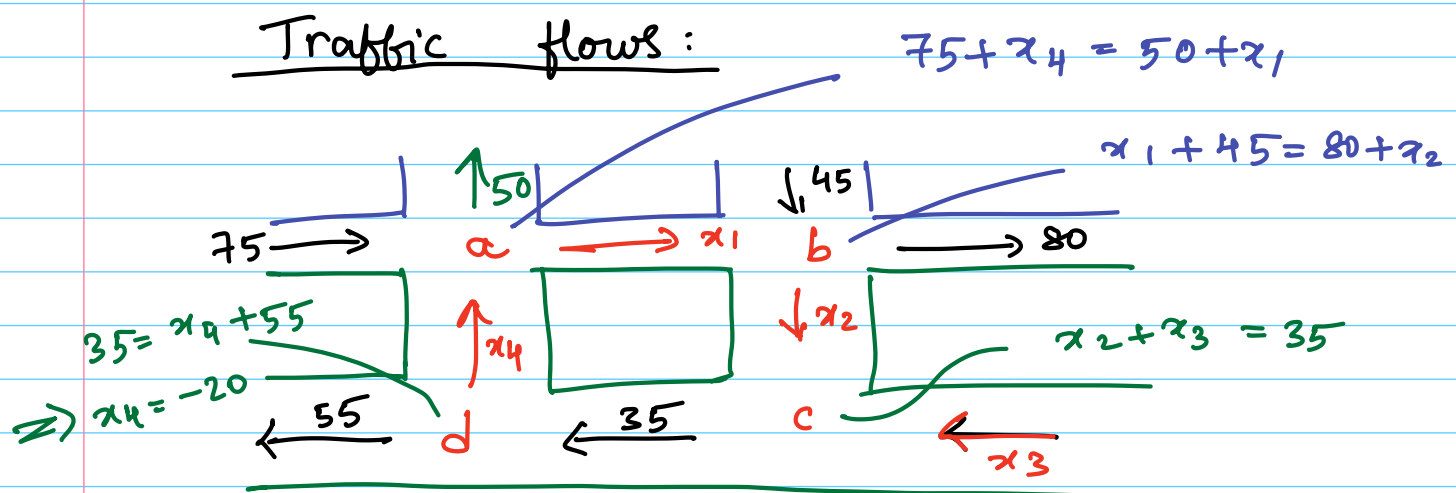
$$= \frac{1}{34} \begin{bmatrix} -80 - 16 \\ -32 - 20 \end{bmatrix}$$

$$= \begin{bmatrix} -96/34 \\ -52/34 \end{bmatrix} = \begin{bmatrix} -48/17 \\ -26/17 \end{bmatrix}$$

$$\therefore i_1 = -\frac{48}{17} \quad i_2 = -\frac{26}{17}$$

mesh currents

Matrices play a very important role in solving large circuits.



Find the traffic flows x_1, x_2, x_3, x_4 .

Principle: For each intersection, incoming traffic = outgoing traffic.

$\therefore$ following eqns. must hold

$$x_1 - x_4 = 25$$

$$x_1 - x_2 = 35$$

$$x_2 + x_3 = 35$$

$$x_4 = -20$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_{\bar{x}} = \underbrace{\begin{bmatrix} 25 \\ 35 \\ 35 \\ -20 \end{bmatrix}}_{\bar{b}}$$

$$\bar{x} = A^{-1} \bar{b}$$

$$\Rightarrow \bar{x} = \begin{bmatrix} 5 \\ -30 \\ +65 \\ -20 \end{bmatrix}$$